



FPT UNIVERSITY

SOFTWARE REQUIREMENTS SPECIFICATION

Course Project Report

WearWhere

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I. Project Overview

1.1. Project Information

- English name: WearWhere
- Vietnamese name: Sàn thương mại điện tử chuyên biệt cho local brand Việt Nam
- Team name: SE1910-NET-Group 4
- Software type: Website and Mobile App

1.2. Project Team

StudentID	Full Name	Role	Email
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HE190769	Ngô Duy Anh	Member	ngoduyanhkover@gmail.com

1.3. Product Background

WearWhere was founded with the mission to elevate the position of Vietnamese local fashion brands in a market currently dominated by mass-market e-commerce platforms. Today, local brands must compete on platforms like Shopee, Lazada, and TikTok Shop - where brand value is drowned in millions of similar products, and success is measured by the ability to offer the deepest discounts rather than design quality or brand storytelling. Market research reveals three critical problems:

1. **Vietnamese Gen Z** (18-30 years old) craves unique style and wants to support Vietnamese designers, but struggles to differentiate quality local brands from factory goods on crowded marketplaces
2. **International tourists** visiting Vietnam seek authentic fashion experiences but cannot discover hidden stores in alleys and face language barriers
3. **Local brand owners** are bearing high commission fees (8-15%), ineffective advertising costs, and losing brand identity when forced to join the price race.

WearWhere is designed as a comprehensive solution for these three target groups. Rather than creating another marketplace, we are building a **premium fashion destination** - where each brand tells its own story through editorial-quality content, users receive style guidance from an AI stylist, and tourists can easily find physical stores through GPS technology. The platform operates seamlessly across both web and mobile app, ensuring a consistent experience whether users are browsing on desktop or shopping on mobile.

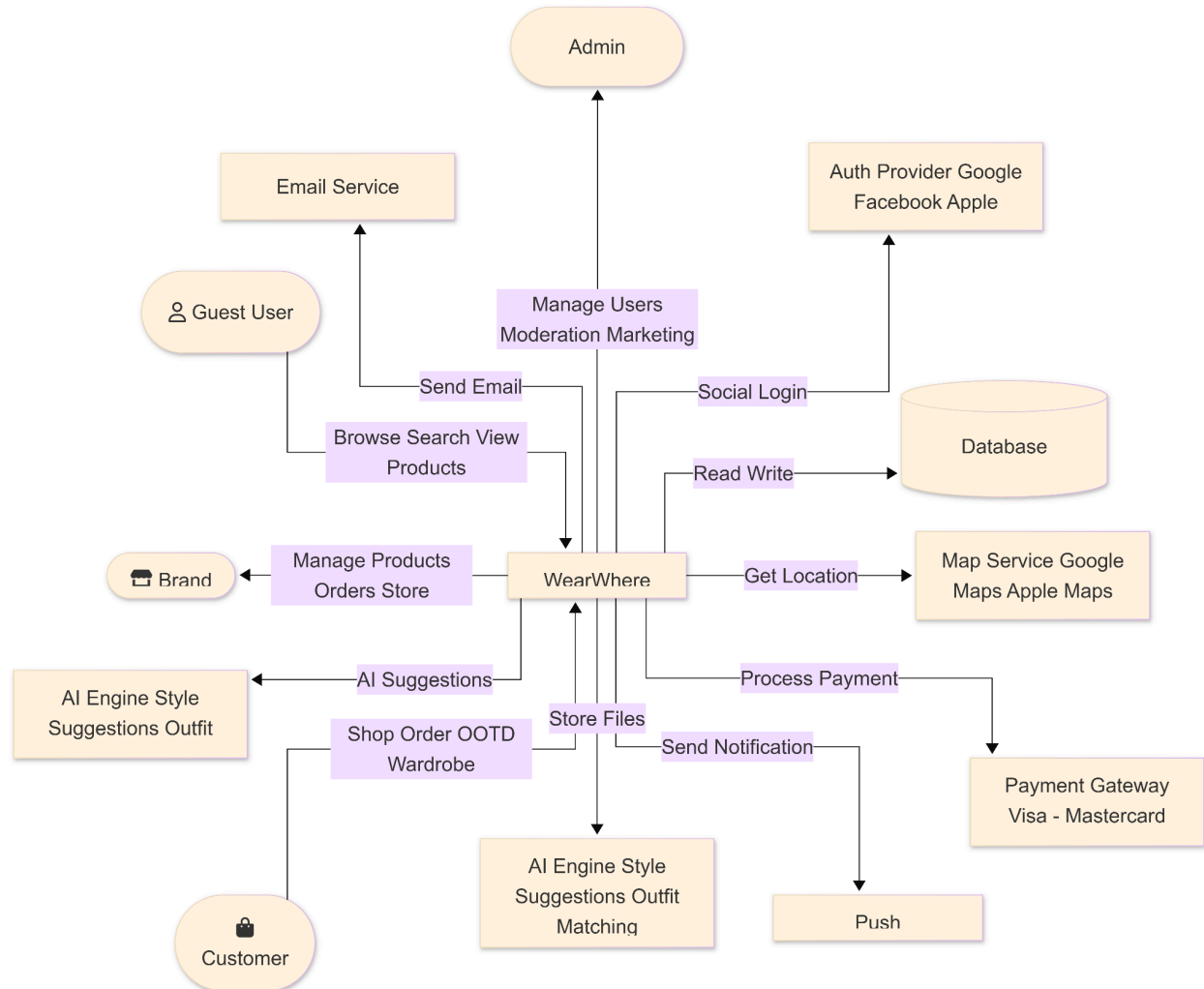
WearWhere's core differentiation lies in combining **discovery technology** (AI recommendations, location-based search, smart wardrobe) with a **brand-first philosophy** (curated selection, storytelling tools, transparent pricing). We don't compete on volume but on quality - each brand must pass verification for design authenticity before onboarding, and the platform design prioritizes inspiration over impulse buying.

1.4. Tools & Infrastructures

Category	Tools / Infrastructure
Frontend - Mobile	Flutter, Dart, Provider/Riverpod
Frontend - Web	Next.js, React, TypeScript, Tailwind CSS
Backend	NestJS, Node.js, TypeScript
Database	PostgreSQL, Redis
Cloud Infrastructure	Google Cloud Platform (GCP)
AI/ML	Gemini API
Maps & Location	Goong Maps API
Payment Gateway	Momo, VNPay
Authentication	Firebase Auth, JWT
Version Control	Git
Project Management	Excel

II. Software Requirement Specification

1. Context Diagram



2. User Requirements

2.1 Actors

#	Actor	Description
1	Guest	People who visit the website but do not use the service
2	Customer	People who log in to the site and use the service
3	Admin	The person who has the right to manage users, full authority on the system
4	Brand Partner	The brand who sell product in the site

2.2 Use Cases

ID	Use Case	Feature	Actor	Use Case Description
Authentication & Profile				

1	Browse Homepage	Navigation	Guest User	Guest users can view homepage and featured brands without logging in
2	View Product Catalog	Product Discovery	Guest User	Guest users can browse products and brand stories without account
3	Register Account	Authentication	Guest User	Guest users can create new account via email/phone/social login
4	Login	Authentication	Guest User	Users can log in via email/phone, Google, Facebook, Apple Sign-In
5	Logout	Authentication	Customer	Users can log out from their account
6	Forgot Password	Authentication	Guest User	Users can reset password via email/SMS OTP
7	Edit Profile	Profile Management	Customer	Users can update personal information, preferences, and avatar
8	Change Password	Profile Management	Customer	Users can change their account password
9	Delete Account	Profile Management	Customer	Users can permanently delete their account and all data
Product Discovery & Shopping				
10	Search Products	Search	Customer	Users can search products by name, brand, style, or category
11	Filter Products	Search	Customer	Users can filter by price range, style, brand, size, color
12	View Product Detail	Product Discovery	Customer	Users can view detailed product info, images, reviews, and size guide
13	View Brand Profile	Brand Discovery	Customer	Users can view brand story, collections, and store locations
14	Add to Cart	Shopping Cart	Customer	Users can add products to shopping cart
15	Update Cart	Shopping Cart	Customer	Users can modify quantity or remove items from cart
16	Add to Wishlist	Wishlist	Customer	Users can save favorite products to wishlist
17	Checkout	Purchase	Customer	Users can proceed to checkout with shipping and payment info
18	Select Payment Method	Payment	Customer	Users can choose COD, Momo, or VNPay
19	Place Order	Purchase	Customer	Users can complete purchase and receive order confirmation
20	Track Order	Order	Customer	Users can view real-time order status and

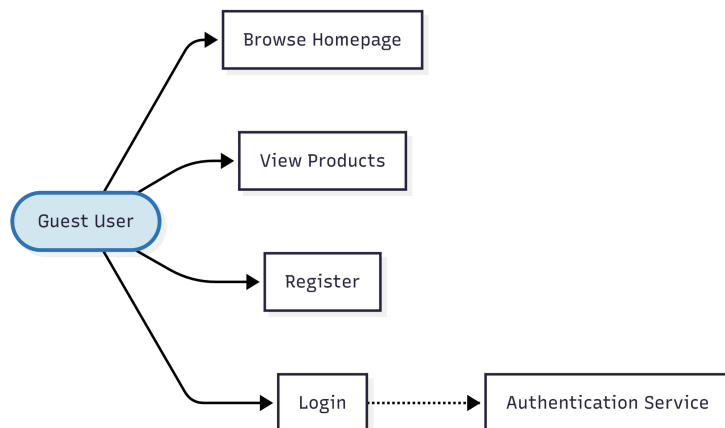
		Management		delivery tracking
21	View Order History	Order Management	Customer	Users can view past orders and reorder items
22	Cancel Order	Order Management	Customer	Users can cancel order before brand confirms
Location & Store Discovery				
23	Find Nearby Stores	Maps & Location	Customer	Users can discover local brand stores near current location
24	Search Store by Area	Maps & Location	Customer	Users can search stores by city/district/street
25	View Store Details	Maps & Location	Customer	Users can view store address, hours, contact, and directions
26	Get Directions	Maps & Location	Customer	Users can get GPS navigation to physical store via Goong Maps
AI & Personalization				
28	Chat with AI Stylist	AI Features	Customer	Users can ask AI chatbot for styling advice and recommendations
29	Get AI Recommendation	AI Features	Customer	Users receive personalized product suggestions based on preferences
30	View Smart Wardrobe	AI Features	Customer	Users can see outfit combinations from purchased items
31	Set Style Preferences	Personalization	Customer	Users can define fashion style preferences during onboarding
Social & Community				
32	Post OOTD	User-Generated Content	Customer	Users can post outfit photos with product tags
33	Like OOTD Post	Social Interaction	Customer	Users can like other users' OOTD posts
34	Comment on OOTD	Social Interaction	Customer	Users can comment on OOTD posts
35	Follow Brand	Social Interaction	Customer	Users can follow favorite brands for updates
36	Follow User	Social Interaction	Customer	Users can follow other fashion enthusiasts
37	Submit Product Review	Reviews	Customer	Users can rate and review purchased products with photos

38	View Reviews	Reviews	Customer	Users can read verified reviews from other shoppers
39	Report Content	Moderation	Customer	Users can report inappropriate content or users
40	Block User	User Management	Customer	Users can block other users from interacting with them
Brand Partner Features				
41	Login to Brand Portal	Brand Authentication	Local Brand Owner	Brand owners access partner portal with credentials
42	Manage Product Catalog	Product Management	Local Brand Owner	Brands can add, edit, or remove products
43	Upload Product Images	Product Management	Local Brand Owner	Brands can upload high-quality product photos
44	Set Product Pricing	Product Management	Local Brand Owner	Brands can set prices and manage inventory
45	View Orders	Order Fulfillment	Local Brand Owner	Brands can view incoming orders
46	Confirm Order	Order Fulfillment	Local Brand Owner	Brands can accept and prepare orders
47	Update Order Status	Order Fulfillment	Local Brand Owner	Brands can update shipping and delivery status
48	Manage Reviews	Review Management	Local Brand Owner	Brands can respond to customer reviews
49	View Analytics Dashboard	Analytics	Local Brand Owner	Brands can view sales, traffic, and customer insights
50	Post Brand Content	Content Management	Local Brand Owner	Brands can publish new collections and brand stories
51	Manage Store Info	Store Management	Local Brand Owner	Brands can update store locations, hours, and contact

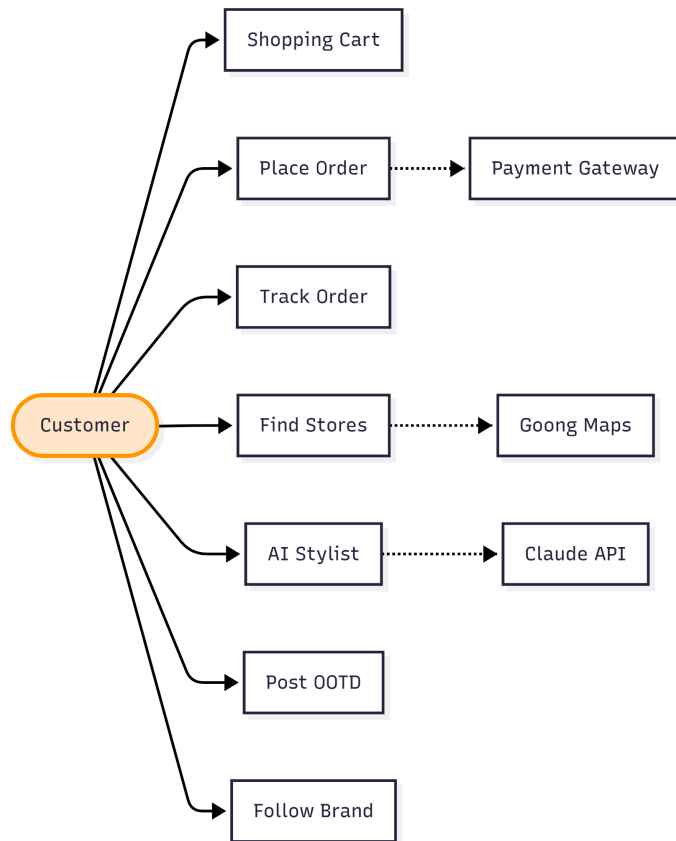
Admin Features				
52	Login to Admin CMS	Admin Authentication	Platform Admin	Admins access CMS with admin credentials
53	Verify Brand Applications	Brand Management	Platform Admin	Admins can approve or reject new brand registrations
54	Manage Users	User Management	Platform Admin	Admins can view, suspend, or delete user accounts
55	Moderate Content	Content Moderation	Platform Admin	Admins can review and remove inappropriate UGC
56	Manage Product Listings	Product Oversight	Platform Admin	Admins can review and remove policy-violating products
57	Handle Reports	Moderation	Platform Admin	Admins can process user reports and take actions
58	Configure Platform Settings	System Configuration	Platform Admin	Admins can adjust commission rates, fees, and policies
59	Manage News & Editorial	Content Management	Platform Admin	Admins can publish fashion news and trend articles
60	View System Analytics	Analytics	Platform Admin	Admins can view GMV, MAU, orders, and platform KPIs
61	Generate Reports	Reporting	Platform Admin	Admins can export revenue, user, and transaction reports
62	Monitor Transactions	Payment Management	Platform Admin	Admins can track payment status and resolve issues

2.3. Diagram(s)

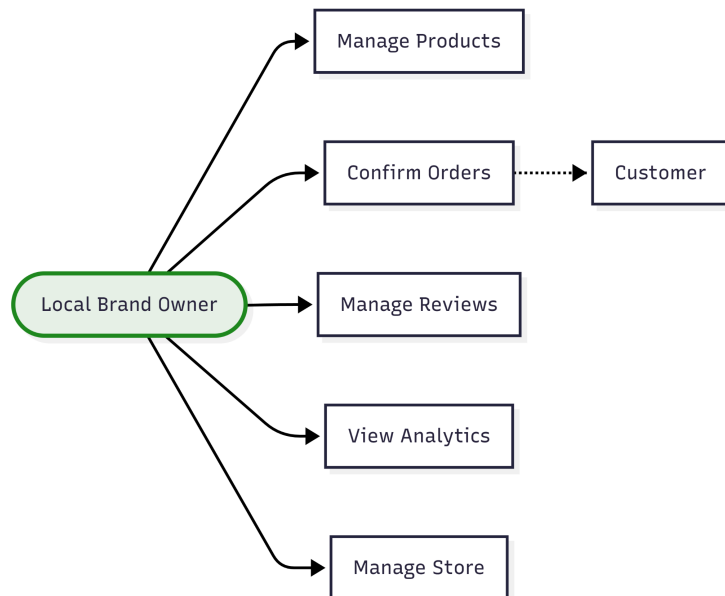
2.3.1. UCs for Guest



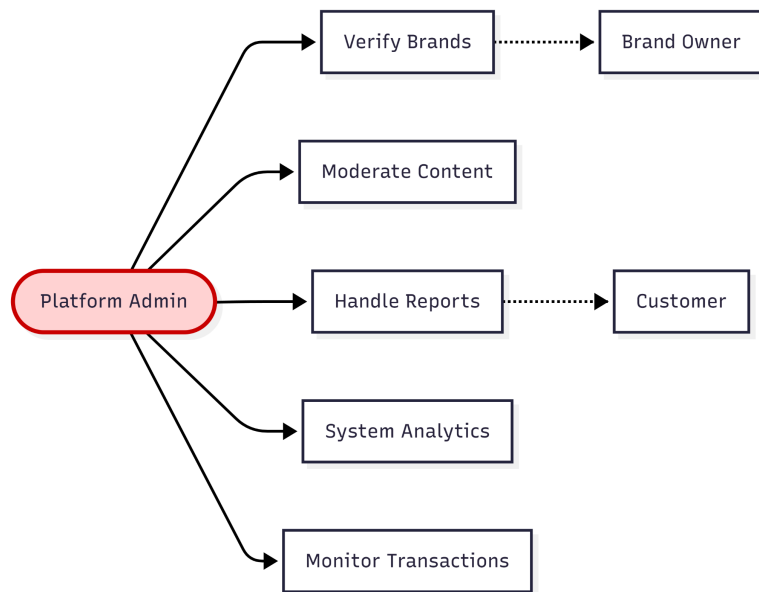
2.3.2. UCs for Customer



2.3.3. UCs for Local Brand Owner

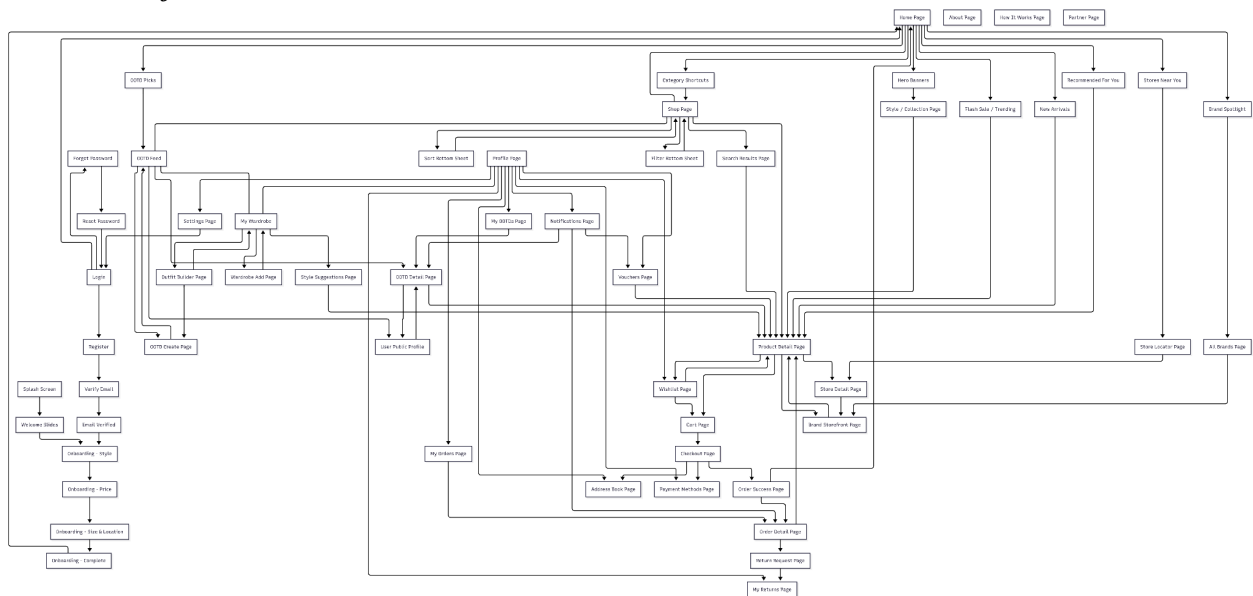


2.3.4. UCs for Platform Admin

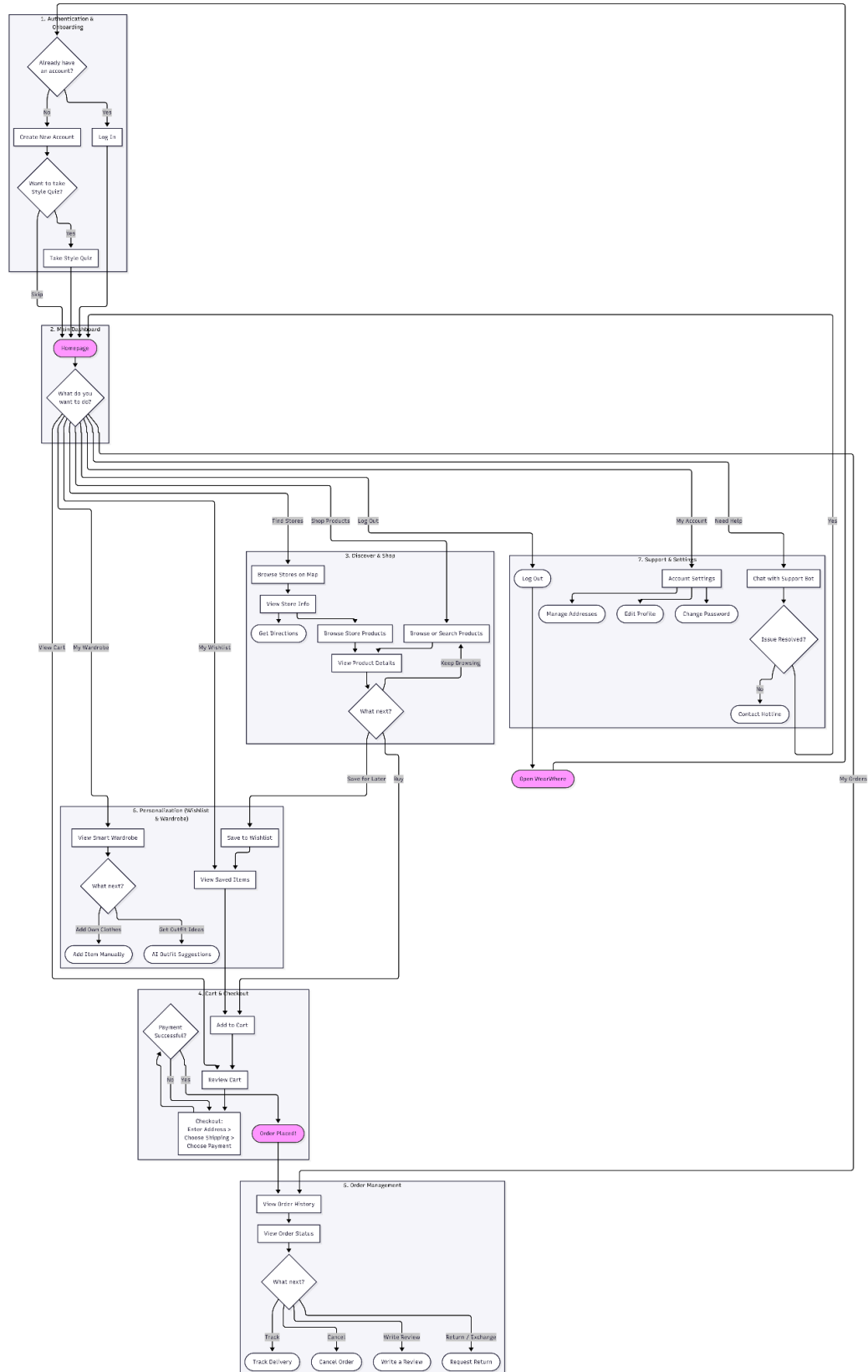


2.4. System Functionality

2.4.1. Screen flows



2.4.2. Customer flows



2.4.1. Screen description

Index	Feature	Screen	Description
1	Splash & Onboarding	Splash Screen	Full screen logo, fade-in animation, 1.5s display
2	Splash & Onboarding	Welcome Slides	3 swipeable intro slides with fashion photography
3	Splash & Onboarding	Onboarding - Style Preferences	Select favorite styles (2x3 grid: Streetwear, Minimalist, Vintage...)
4	Splash & Onboarding	Onboarding - Price Range	Select shopping budget (range slider VND)
5	Splash & Onboarding	Onboarding - Size & Location	Select clothing size, shoe size, gender, city
6	Splash & Onboarding	Onboarding - Complete	Onboarding completion screen, success animation
7	Auth	Login	Email/password login + social login (Google, Apple, Facebook)
8	Auth	Register	New account registration, password strength indicator
9	Auth	Forgot Password	Enter email to receive reset password link
10	Auth	Reset Password	Set new password
11	Auth	Verify Email	Enter 6-digit OTP code for email verification
12	Auth	Email Verified	Email verification success, redirect to onboarding
13	Home	Home Page	Homepage: hero banner, categories, trending, new arrivals, flash sale, brand spotlight, OOTD picks, nearby stores
14	Shop	Shop Page	Browse products: search bar, category tabs, filter/sort bar, 2-column product grid
15	Shop	Filter Bottom Sheet	Filters: price range, brands, sizes, colors, style tags, rating
16	Shop	Sort Bottom Sheet	Sort options: relevance, price, newest, popular, rating
17	Shop	Search Results Page	Search results: recent searches, suggestions, results grid
18	Shop	Product Detail Page	Product detail: image gallery, info, color/size selector, reviews, related products, store availability
19	Shop	Style/Collection Page	Style page by slug: banner + curated products
20	Shop	All Brands Page	A-Z brand listing, search, brand cards grid
21	Shop	Brand Storefront Page	Brand page: banner, logo, follow, category tabs,

			product grid
22	Shop	Vouchers Page	Vouchers: available/saved tabs, voucher cards, detail bottom sheet
23	OOTD	OOTD Feed Page	Instagram-style social fashion feed: post cards, like/comment/share
24	OOTD	OOTD Create Page	Create post: select photo, tag products, caption, style tags, preview
25	OOTD	OOTD Detail Page	Post detail: full image, tagged products, comments thread
26	OOTD	User Public Profile Page	Public profile: avatar, bio, stats, posts grid, follow/unfollow
27	Wardrobe	My Wardrobe Page	Virtual wardrobe: stats, category filter, item grid, sort, FAB add
28	Wardrobe	Wardrobe Add Page	Add item: camera/gallery, category, name, brand, color, size
29	Wardrobe	Outfit Builder Page	Build outfit: canvas drop zones, item picker, save/share
30	Wardrobe	Style Suggestions Page	AI outfit suggestions: card stack swipe, shop missing piece CTA
31	Account	Profile Page	Profile overview: avatar, name, bio, stats, quick links grid
32	Account	My Orders Page	Order list: tabs (All/Processing/Shipped/Delivered/Cancelled)
33	Account	Order Detail Page	Order detail: status timeline, items, shipping, payment summary
34	Account	Return Request Page	Return request: select items, reason, upload photos, notes
35	Account	My Returns Page	Returns list: cards with status badge
36	Account	My OOTDs Page	Grid of user's OOTD posts, edit/delete
37	Account	Wishlist Page	Wishlist: product grid, remove, move to cart
38	Account	Notifications Page	Notifications: tabs All/Orders/Social/Promotions, read/unread
39	Account	Address Book Page	Address book: list, add/edit form, default toggle
40	Account	Payment Methods Page	Payment methods: saved cards, e-wallets, add new
41	Account	Settings Page	Settings: language, notifications, privacy, theme, about, logout, delete account
42	Cart & Checkout	Cart/Checkout Page	Cart + checkout: item list, voucher, shipping, payment, place order

43	Cart & Checkout	Order Success Page	Order success: animation, order number, estimated delivery
44	Store Locator	Store Locator Page	Store map: map view, search, bottom sheet list, list view toggle
45	Store Locator	Store Detail Page	Store detail: photos, address, hours, brands, get directions
46	Info	About Page	About WearWhere
47	Info	How It Works Page	How it works guide
48	Info	Partner Page	Become a seller/partner registration
49	Brand Portal	Brand Dashboard	Brand overview: statistics, charts
50	Brand Portal	Brand Sales Page	Brand sales report
51	Brand Portal	Brand Traffic Page	Brand traffic/visits report
52	Brand Portal	Brand Profile Page	Manage brand profile information
53	Brand Portal	Brand Locations Page	Brand store locations list
54	Brand Portal	Brand Location Form Page	Add/edit brand store location
55	Brand Portal	Brand Products Page	Brand product listing
56	Brand Portal	Brand Product Form Page	Add/edit product
57	Brand Portal	Brand Categories Page	Manage brand product categories
58	Brand Portal	Brand Orders Page	Brand order listing
59	Brand Portal	Brand Order Detail Page	Brand order detail
60	Brand Portal	Brand Returns Page	Brand returns listing
61	Brand Portal	Brand Return Detail Page	Return request detail
62	Brand Portal	Brand Settings Page	Brand account settings
63	Admin Portal	Admin Dashboard	Admin overview: platform statistics
64	Admin Portal	Admin Monitor Page	Real-time system monitoring
65	Admin - Users	Admin Users Page	Manage user listing
66	Admin - Users	Admin User Detail Page	User detail and actions
67	Admin - Users	Admin Deleted Accounts Page	Deleted accounts listing
68	Admin - Products	Admin Products Page	Manage all products
69	Admin - Products	Admin Reported	Reported products listing

		Products Page	
70	Admin - Products	Admin Categories Page	Manage product categories
71	Admin - Products	Admin Style Tags Page	Manage style tags
72	Admin - Brands	Admin Brands Page	Manage brand listing
73	Admin - Brands	Admin Brand Applications Page	New brand applications
74	Admin - Brands	Admin Review Application Page	Review brand application
75	Admin - Brands	Admin Brand Detail Page	Brand detail (admin view)
76	Admin - Brands	Admin Brand Subscriptions Page	Manage brand subscription plans
77	Admin - Brands	Admin Verify Brand Page	Verify brand
78	Admin - Brands	Admin Suspend Brand Page	Suspend brand
79	Admin - Moderation	Admin Moderation Queue Page	Content moderation queue
80	Admin - Moderation	Admin Review Reported OOTD Page	Review reported OOTD
81	Admin - Moderation	Admin Auto Filter Settings Page	Auto filter settings
82	Admin - Moderation	Admin Keyword Blacklist Page	Manage blocked keywords
83	Admin - Orders	Admin All Orders Page	All platform orders
84	Admin - Orders	Admin Order Detail Page	Order detail (admin view)
85	Admin - Orders	Admin Disputes Page	Disputes listing
86	Admin - Orders	Admin Resolve Dispute Page	Resolve dispute
87	Admin - Orders	Admin Refund Management Page	Manage refunds
88	Admin - Marketing	Admin Platform Promotions Page	Manage platform promotions
89	Admin - Marketing	Admin Create Voucher Page	Create new voucher
90	Admin - Marketing	Admin Push Notifications Page	Manage push notifications

91	Admin - Marketing	Admin Email Campaigns Page	Manage email campaigns
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2.4.4. Screen authorization

Index	Screen	Guest	Customer	Brand	Admin
1	Splash Screen	X	X	X	X
2	Welcome Slides	X	X	X	X
3	Onboarding - Style Preferences	X	X	X	X
4	Onboarding - Price Range	X	X	X	X
5	Onboarding - Size & Location	X	X	X	X
6	Onboarding - Complete	X	X	X	X
7	Login	X	X	X	X
8	Register	X	X	X	X
9	Forgot Password	X	X	X	X
10	Reset Password	X	X	X	X
11	Verify Email	X	X	X	X
12	Email Verified	X	X	X	X
13	Home Page	X	X		
14	Shop Page	X	X		
15	Filter Bottom Sheet	X	X		
16	Sort Bottom Sheet	X	X		
17	Search Results Page	X	X		
18	Product Detail Page	X	X		
19	Style/Collection Page	X	X		
20	All Brands Page	X	X		
21	Brand Storefront Page	X	X		
22	Vouchers Page		X		
23	OOTD Feed Page	X	X		
24	OOTD Create Page		X		
25	OOTD Detail Page	X	X		
26	User Public Profile Page	X	X		
27	My Wardrobe Page		X		

28	Wardrobe Add Page		X		
29	Outfit Builder Page		X		
30	Style Suggestions Page		X		
31	Profile Page		X		
32	My Orders Page		X		
33	Order Detail Page		X		
34	Return Request Page		X		
35	My Returns Page		X		
36	My OOTDs Page		X		
37	Wishlist Page		X		
38	Notifications Page		X		
39	Address Book Page		X		
40	Payment Methods Page		X		
41	Settings Page		X		
42	Cart/Checkout Page		X		
43	Order Success Page		X		
44	Store Locator Page	X	X		
45	Store Detail Page	X	X		
46	About Page	X	X		
47	How It Works Page	X	X		
48	Partner Page	X	X		
49	Brand Dashboard			X	
50	Brand Sales Page			X	
51	Brand Traffic Page			X	
52	Brand Profile Page			X	
53	Brand Locations Page			X	
54	Brand Location Form Page			X	
55	Brand Products Page			X	
56	Brand Product Form Page			X	
57	Brand Categories Page			X	
58	Brand Orders Page			X	
59	Brand Order Detail Page			X	

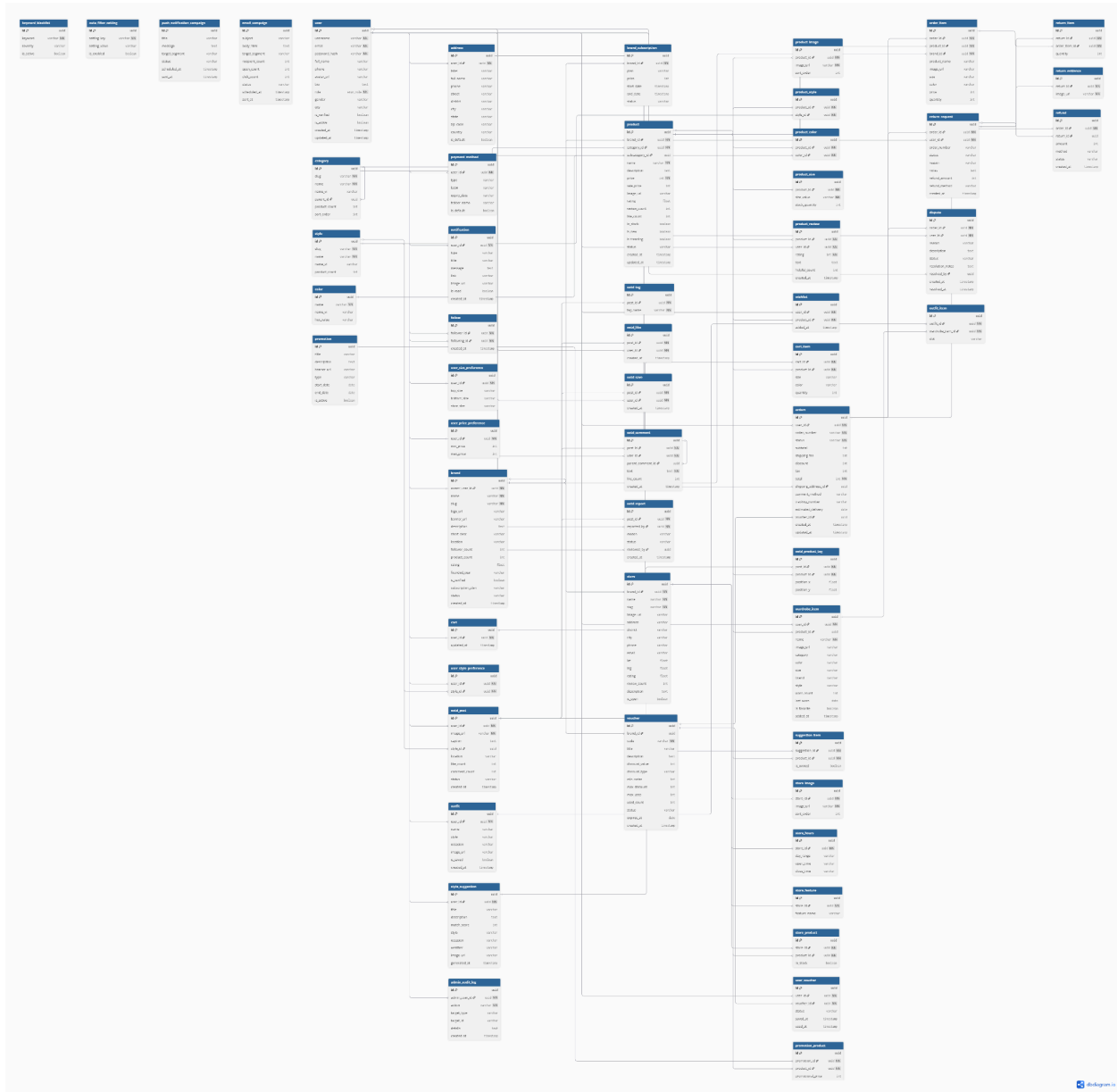
60	Brand Returns Page			X	
61	Brand Return Detail Page			X	
62	Brand Settings Page			X	
63	Admin Dashboard				X
64	Admin Monitor Page				X
65	Admin Users Page				X
66	Admin User Detail Page				X
67	Admin Deleted Accounts Page				X
68	Admin Products Page				X
69	Admin Reported Products Page				X
70	Admin Categories Page				X
71	Admin Style Tags Page				X
72	Admin Brands Page				X
73	Admin Brand Applications Page				X
74	Admin Review Application Page				X
75	Admin Brand Detail Page				X
76	Admin Brand Subscriptions Page				X
77	Admin Verify Brand Page				X
78	Admin Suspend Brand Page				X
79	Admin Moderation Queue Page				X
80	Admin Review Reported OOTD Page				X
81	Admin Auto Filter Settings Page				X
82	Admin Keyword Blacklist Page				X
83	Admin All Orders Page				X
84	Admin Order Detail Page				X
85	Admin Disputes Page				X
86	Admin Resolve Dispute Page				X
87	Admin Refund Management Page				X
88	Admin Platform Promotions Page				X
89	Admin Create Voucher Page				X
90	Admin Push Notifications Page				X
91	Admin Email Campaigns Page				X

2.4.5. Non-UI Functions

Index	Category	Feature	System Function	Description
1	Performance	Page Load	Web & Mobile	Product pages load in < 2 seconds
2	Performance	API Response	Backend API	95% of requests respond in < 500ms
3	Performance	AI Response	AI Stylist	AI generates responses within 3 seconds
4	Performance	Concurrent Users	Infrastructure	Support 10,000 simultaneous users
5	Reliability	System Uptime	All Systems	99.9% availability
6	Reliability	Data Backup	Database	Daily automated backups with 30-day retention
7	Reliability	Recovery Time	Infrastructure	RTO < 4 hours
8	Usability	Navigation	UI/UX	Max 3 clicks to reach any product
9	Usability	Onboarding	Mobile & Web	Style quiz completes in < 2 minutes
10	Usability	Accessibility	All Interfaces	WCAG 2.1 AA compliance
11	Security	Password Security	Authentication	bcrypt hashing with cost factor 12
12	Security	Session Management	Authentication	JWT tokens with 30-day expiration
13	Security	Access Control	Authorization	Role-based permissions (Guest, Customer, Brand, Admin)
14	Security	Data Encryption	All Systems	HTTPS/TLS 1.3 for all connections
15	Security	Payment Security	Payment Gateway	PCI-DSS compliant, no card data stored
16	Security	Brute Force Protection	Authentication	5 failed logins = 15min lockout
17	Security	Audit Logging	Admin Panel	All admin actions logged with timestamp
18	Scalability	Rate Limiting	API	100 requests/minute per user
19	Scalability	CDN	Static Assets	Global CDN for images and files

20	Compatibility	Browser Support	Web App	Chrome 90+, Safari 14+, Firefox 88+, Edge 90+
21	Compatibility	Mobile OS	Mobile App	iOS 14+ and Android 10+
22	Compliance	Apple Sign-In	iOS App	Required per App Store Guideline 4.8
23	Compliance	Account Deletion	User Management	Users can delete accounts and all data
24	Compliance	Content Moderation	UGC System	Moderation queue for OOTD posts and reviews
25	Compliance	GDPR	Data Privacy	Data deletion within 90 days upon request

2.5. Entity Relationship Diagram



3. Use Case Specifications

3.1. Authentication & Profile

3.1.1. UC Browse Homepage

ID and Name	UC01 - Browse Homepage		
Created by	Bùi Anh Linh		
Primary Actor	Guest User	Secondary Actor	None

Description	Guest users view homepage with featured brands, trending products, and OOTD feed without logging in.
Preconditions	User has internet connection and accesses WearWhere web/app.
Main Sequence	1. Guest opens WearWhere 2. System displays homepage with featured content 3. Guest browses freely without account
Alternative Sequences	None
Postconditions	Guest can explore platform and decide to register or continue browsing.
Business Rule	- Limited features without login (no cart save, wishlist, OOTD posting)

3.1.2. UC View Product Catalog

ID and Name	UC02 - View Product Catalog		
Created by	Bùi Anh Linh		
Primary Actor	Guest User	Secondary Actor	None
Description	Guest users browse products and brand stories without creating an account.		
Preconditions	User accesses product catalog page.		
Main Sequence	1. Guest navigates to products section 2. System displays product grid with filters 3. Guest views products, images, and basic info		
Alternative Sequences	If guest clicks wishlist/cart: System prompts login.		
Postconditions	Guest understands product offerings and may proceed to register.		
Business Rule	- Full product details visible to guests - Prices shown with clear currency		

3.1.3. UC Register Account

ID and Name	UC03 - Register Account		
Created by	Bùi Anh Linh		
Primary Actor	Guest User	Secondary Actor	Authentication Service
Description	Guest users create account via email/phone or social login (Google, Facebook, Apple).		
Preconditions	1. User not logged in 2. Valid email/phone available		
Main Sequence	1. Guest clicks Register 2. System shows registration form 3. Guest enters credentials or selects social login 4. System validates and creates account 5. System sends verification email/SMS 6. Guest confirms and logs in		
Alternative Sequences	3a. Email exists: Show login link 5a. Verification fails: Resend option		
Postconditions	User account created. User logged in with valid session.		
Business Rule	- Password min 8 chars - Email/phone must be unique - Apple Sign-In required on iOS if other social logins exist		

3.1.4. UC Login

ID and Name	UC04 - Login		
Created by	Bùi Anh Linh		
Primary Actor	Guest User	Secondary Actor	Authentication Service
Description	Users log in with email/phone, Google, Facebook, or Apple Sign-In.		
Preconditions	1. User has registered account 2. Not currently logged in		
Main Sequence	1. Guest clicks Login 2. System shows login options 3. Guest enters credentials 4. System validates and creates session 5. System redirects to homepage		

Alternative Sequences	4a. Wrong password: Error message, retry (max 5 attempts) 4b. Account not found: Suggest registration 4c. Network error: Retry option
Postconditions	User authenticated with session token. Full features accessible.
Business Rule	- Session expires after 30 days inactivity - 5 failed attempts = 15min lockout

3.1.5. UC Logout

ID and Name	UC05 - Logout		
Created by	Bùi Anh Linh		
Primary Actor	Customer	Secondary Actor	None
Description	Users log out from their account.		
Preconditions	User is logged in.		
Main Sequence	1. Customer clicks Logout 2. System invalidates session token 3. System redirects to homepage as guest		
Alternative Sequences	None		
Postconditions	User logged out. Session token deleted. Guest permissions only.		
Business Rule	- Unsaved cart items retained for 7 days if user logs back in		

3.1.6. UC Forget Password

ID and Name	UC06 - Forgot Password		
Created by	Bùi Anh Linh		
Primary Actor	Guest User	Secondary Actor	Email/SMS Service
Description	Users reset password via email/SMS OTP.		

Preconditions	User has registered account but forgot password.
Main Sequence	1. Guest clicks Forgot Password 2. Guest enters email/phone 3. System sends OTP code 4. Guest enters OTP 5. Guest sets new password 6. System updates password and logs in
Alternative Sequences	3a. Email/phone not found: Show error 4a. Wrong OTP: Retry (3 attempts) 4b. OTP expired (15min): Resend option
Postconditions	Password updated. User logged in automatically.
Business Rule	- OTP valid for 15 minutes - Max 3 OTP requests per hour

3.1.7. UC Edit Profile

ID and Name	UC07 - Edit Profile		
Created by	Bùi Anh Linh		
Primary Actor	Customer	Secondary Actor	None
Description	Users update personal information, preferences, and avatar.		
Preconditions	User is logged in.		
Main Sequence	1. Customer navigates to Profile 2. Customer clicks Edit 3. Customer updates fields (name, avatar, preferences) 4. Customer saves changes 5. System validates and updates profile		
Alternative Sequences	5a. Invalid data: Show field-specific errors 5b. Upload image > 5MB: Show size limit error		
Postconditions	Profile updated. Changes reflected across platform.		
Business Rule	- Name required - Avatar max 5MB - Style preferences affect AI recommendations		

3.1.8. UC Change Password

ID and Name	UC08 - Change Password		
Created by	Bùi Anh Linh		
Primary Actor	Customer	Secondary Actor	None
Description	Users change their account password.		
Preconditions	User is logged in.		
Main Sequence	1. Customer navigates to Security settings 2. Customer enters current password 3. Customer enters new password (2x) 4. System validates and updates password 5. System logs out all devices except current		
Alternative Sequences	2a. Wrong current password: Error and retry 3a. New passwords don't match: Error 3b. Weak password: Show strength requirements		
Postconditions	Password updated. All other sessions invalidated.		
Business Rule	- New password must differ from last 3 passwords - Min 8 chars, 1 number, 1 special char		

3.1.9. UC Delete Account

ID and Name	UC09 - Delete Account		
Created by	Bùi Anh Linh		
Primary Actor	Customer	Secondary Actor	None
Description	Users permanently delete account and all associated data.		
Preconditions	1. User is logged in 2. No pending orders		

Main Sequence	1. Customer navigates to Account Settings 2. Customer clicks Delete Account 3. System shows warning dialog 4. Customer confirms with password 5. System deletes all user data 6. System sends confirmation email 7. System logs out user
Alternative Sequences	4a. Wrong password: Retry 4b. Pending orders exist: Show orders, block deletion
Postconditions	Account deleted within 90 days. User logged out. Data anonymized.
Business Rule	- GDPR/compliance: Data deleted in 90 days - Order history retained anonymously for accounting - Cannot be undone

3.2. Product Discovery & Shopping

3.2.1. UC Search Products

ID and Name	UC10 - Search Products		
Created by	Ngo Duy Anh		
Primary Actor	Customer	Secondary Actor	None
Description	Users search products by name, brand, style, or category.		
Preconditions	User on platform (guest or logged in).		
Main Sequence	1. User enters search query 2. System searches product database 3. System displays matching results with filters 4. User refines search or selects product		
Alternative Sequences	2a. No results: Show trending products instead 2b. Typo detected: Suggest correct spelling		
Postconditions	Relevant products displayed. User can proceed to view details.		
Business Rule	- Search indexes product name, brand, description, tags - Results sorted by relevance + popularity		

3.2.2. UC Filter Products

ID and Name	UC11 - Filter Products		
Created by	Ngo Duy Anh		
Primary Actor	Customer	Secondary Actor	None
Description	Users filter products by price, style, brand, size, color.		
Preconditions	User viewing product catalog or search results.		
Main Sequence	1. User opens filter panel 2. User selects filter criteria 3. System applies filters and updates results 4. User views filtered products		
Alternative Sequences	3a. No products match filters: Show message and suggest removing filters		
Postconditions	Product list filtered. User can further refine or view products.		
Business Rule	- Multiple filters combine with AND logic - Filter count shown on button		

3.2.3. UC View Product Detail

ID and Name	UC12 - View Product Detail		
Created by	Ngo Duy Anh		
Primary Actor	Customer	Secondary Actor	None
Description	Users view detailed product info including images, reviews, size guide, and stock.		
Preconditions	User clicks on a product.		
Main Sequence	1. User clicks product card 2. System displays product detail page 3. User views images, description, price, sizes, reviews 4. User can add to cart or wishlist		
Alternative Sequences	2a. Product out of stock: Show 'Notify when available' button		
Postconditions	User has full product information to make purchase decision.		

Business Rule	- Show real-time stock - Display verified purchase reviews first
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3.2.4. UC View Brand Profile

ID and Name	UC13 - View Brand Profile		
Created by	Ngo Duy Anh		
Primary Actor	Customer	Secondary Actor	None
Description	Users view brand story, collections, and physical store locations.		
Preconditions	User clicks on brand name or logo.		
Main Sequence	1. User navigates to brand profile 2. System displays brand page with story, products, stores 3. User explores brand content and products		
Alternative Sequences	None		
Postconditions	User understands brand identity and can follow or shop.		
Business Rule	- Show all products from brand - Store locations with map integration		

3.2.5. UC Add to Cart

ID and Name	UC14 - Add to Cart		
Created by	Ngo Duy Anh		
Primary Actor	Customer	Secondary Actor	None
Description	Users add products with selected size/color to shopping cart.		
Preconditions	1. User logged in 2. Product in stock		

Main Sequence	1. User selects size/color variant 2. User clicks Add to Cart 3. System validates selection and stock 4. System adds item to cart 5. System shows cart preview popup
Alternative Sequences	3a. No size selected: Prompt to select 3b. Out of stock: Show error and 'Notify' option 3c. Already in cart: Update quantity
Postconditions	Item in cart. Cart count updated. User can continue shopping or checkout.
Business Rule	- Cart persists across sessions - Stock reserved for 30 minutes after adding

3.2.6. UC Update Cart

ID and Name	UC15 - Update Cart		
Created by	Ngo Duy Anh		
Primary Actor	Customer	Secondary Actor	None
Description	Users modify quantity or remove items from shopping cart.		
Preconditions	User has items in cart.		
Main Sequence	1. User opens cart 2. User adjusts quantity or clicks remove 3. System updates cart and recalculates total 4. System shows updated cart		
Alternative Sequences	3a. Quantity exceeds stock: Show max available and adjust 3b. All items removed: Show empty cart message		
Postconditions	Cart reflects changes. Total recalculated. Stock reservations updated.		
Business Rule	- Max quantity per item: 10 - If stock drops below cart quantity, alert user		

3.2.7. UC Add to Wishlist

ID and Name	UC16 - Add to Wishlist
Created by	Ngo Duy Anh

Primary Actor	Customer	Secondary Actor	None
Description	Users save favorite products to wishlist for later.		
Preconditions	User logged in.		
Main Sequence	1. User clicks heart icon on product 2. System adds product to wishlist 3. System shows confirmation animation		
Alternative Sequences	2a. Already in wishlist: Remove from wishlist instead		
Postconditions	Product saved to wishlist. User can access via Wishlist page.		
Business Rule	- Wishlist unlimited items - Notify user if wishlist item goes on sale or out of stock		

3.2.8. UC Checkout

ID and Name	UC17 - Checkout		
Created by	Ngo Duy Anh		
Primary Actor	Customer	Secondary Actor	None
Description	Users proceed to checkout with shipping and payment information.		
Preconditions	1. User has items in cart 2. Logged in		
Main Sequence	1. User clicks Checkout 2. System shows checkout page with cart summary 3. User confirms/edits shipping address 4. System calculates shipping fees 5. User proceeds to payment selection		
Alternative Sequences	3a. No saved address: Prompt to add address 4a. Items from multiple brands: Show separate shipping fees per brand		
Postconditions	User ready to select payment method and place order.		
Business Rule	- Min order value: 50,000 VND - Shipping calculated per brand		

3.2.9. UC Select Payment Method

ID and Name	UC18 - Select Payment Method		
Created by	Ngo Duy Anh		
Primary Actor	Customer	Secondary Actor	None
Description	Users choose payment method: COD, Momo, or VNPay.		
Preconditions	User at payment step in checkout.		
Main Sequence	1. System displays payment options 2. User selects payment method 3. System shows relevant instructions/fees 4. User confirms selection		
Alternative Sequences	None		
Postconditions	Payment method selected. User ready to place order.		
Business Rule	- COD available all orders - Online payment preferred (lower brand fee)		

3.2.10. UC Place Order

ID and Name	UC19 - Place Order		
Created by	Ngo Duy Anh		
Primary Actor	Customer	Secondary Actor	Payment Gateway, Brand Partner
Description	Users complete purchase. System processes payment and notifies brands.		
Preconditions	1. Cart validated 2. Shipping address confirmed 3. Payment method selected		
Main Sequence	1. User clicks Place Order 2. System validates order 3. IF online payment: Redirect to gateway, process payment 4. System creates order with unique ID 5. System sends confirmation to user 6. System notifies brands to prepare shipment		

Alternative Sequences	3a. Payment fails: Show error, return to payment selection 2a. Stock depleted: Remove item, show message
Postconditions	Order created. User receives confirmation. Brands receive order notification. Cart cleared.
Business Rule	- Commission (8-12%) deducted from brand payout - COD orders cancellable until brand confirms

3.2.11. UC Track Order

ID and Name	UC20 - Track Order		
Created by	Ngo Duy Anh		
Primary Actor	Customer	Secondary Actor	None
Description	Users view real-time order status and delivery tracking.		
Preconditions	User has placed order.		
Main Sequence	1. User navigates to My Orders 2. User selects specific order 3. System displays order status timeline 4. User views current status and estimated delivery		
Alternative Sequences	None		
Postconditions	User informed of order progress.		
Business Rule	- Status: Pending → Confirmed → Preparing → Shipped → Delivered - Update within 1 hour of brand action		

3.2.12. UC View Order History

ID and Name	UC21 - View Order History		
Created by	Ngo Duy Anh		
Primary Actor	Customer	Secondary Actor	None
Description	Users view past orders and reorder items easily.		
Preconditions	1. User logged in 2. Has order history		

Main Sequence	1. User navigates to Order History 2. System displays all past orders 3. User filters/searches orders 4. User clicks order to view details or reorder
Alternative Sequences	None
Postconditions	User can review past purchases or quickly reorder.
Business Rule	- Show orders from last 2 years - Reorder adds items to cart with current pricing

3.2.13. UC Cancel Order

ID and Name	UC22 - Cancel Order		
Created by	Ngo Duy Anh		
Primary Actor	Customer	Secondary Actor	Brand Partner
Description	Users cancel order before brand confirms shipment.		
Preconditions	Order status is Pending or Confirmed (not yet shipped).		
Main Sequence	1. User opens order details 2. User clicks Cancel Order 3. System shows confirmation dialog 4. User confirms cancellation 5. System cancels order and processes refund (if paid) 6. System notifies brand of cancellation		
Alternative Sequences	1a. Order already shipped: Cancel button disabled, show contact support		
Postconditions	Order cancelled. Refund processed (if applicable). Brand notified.		
Business Rule	- COD: Cancel anytime before shipped - Paid orders: Cancel within 1 hour for full refund, after 1 hour requires brand approval		

3.3. Location & Store Discovery

3.3.1. UC Find Nearby Stores

ID and Name	UC23 - Find Nearby Stores		
Created by	Bùi Anh Linh		
Primary Actor	Customer	Secondary Actor	Goong Maps API

Description	Users discover local brand stores near their current GPS location.
Preconditions	1. User grants location permission 2. GPS enabled
Main Sequence	1. User opens Map/Store Finder 2. System requests location permission 3. User grants permission 4. System fetches nearby stores (5km radius) 5. System displays stores on map with pins
Alternative Sequences	3a. Permission denied: Prompt manual location entry 4a. No stores nearby: Expand radius to 10km
Postconditions	User sees nearby stores and can navigate to them.
Business Rule	- Use Goong Maps API - Show store distance, ratings, and open status

3.3.2. UC Search Store by Area

ID and Name	UC24 - Search Store by Area		
Created by	Bùi Anh Linh		
Primary Actor	Customer	Secondary Actor	None
Description	Users search stores by city, district, or street name.		
Preconditions	User on map/store finder page.		
Main Sequence	1. User enters location query 2. System searches store database by address 3. System displays matching stores on map 4. User selects store for details		
Alternative Sequences	2a. No results: Suggest nearby areas		
Postconditions	Relevant stores displayed. User can view details or get directions.		
Business Rule	- Search by partial address - Show popular shopping districts first		

3.3.3. UC View Store Details

ID and Name	UC25 - View Store Details
Created by	Bùi Anh Linh

Primary Actor	Customer	Secondary Actor	None
Description	Users view store address, hours, contact, photos, and directions.		
Preconditions	User clicks on store pin or list item.		
Main Sequence	1. User selects store 2. System displays store detail card 3. User views address, hours, phone, photos 4. User can call, get directions, or view products		
Alternative Sequences	None		
Postconditions	User has full store information to visit in person.		
Business Rule	- Show current open/closed status - Display store photos from brand profile		

3.3.4. UC Get Directions

ID and Name	UC26 - Get Directions		
Created by	Bùi Anh Linh		
Primary Actor	Customer	Secondary Actor	Goong Maps API
Description	Users get GPS navigation to physical store via Goong Maps.		
Preconditions	1. User viewing store details 2. Location permission granted		
Main Sequence	1. User clicks Get Directions 2. System calculates route using Goong Maps 3. System displays turn-by-turn navigation 4. User follows directions to store		
Alternative Sequences	1a. Location off: Prompt to enable 2a. Route unavailable: Show manual address		
Postconditions	User navigating to store with live directions.		
Business Rule	- Use Goong Maps for Vietnam accuracy - Show estimated travel time and distance		

3.4. AI & Personalization

3.4.1. UC Chat with AI Stylist

ID and Name	UC28 - Chat with AI Stylist		
Created by	Bùi Anh Linh		
Primary Actor	Customer	Secondary Actor	Claude API
Description	Users ask AI chatbot for styling advice, product recommendations, and fashion questions.		
Preconditions	User logged in.		
Main Sequence	1. User opens AI Stylist chat 2. User types fashion question 3. System sends query to Claude API 4. AI generates personalized response 5. System displays answer with product links		
Alternative Sequences	4a. API timeout: Show retry option 4b. Inappropriate query: Polite decline with guidelines		
Postconditions	User receives styling advice. May click products to view/purchase.		
Business Rule	- Track token usage - Free tier limits apply - Responses reference WearWhere catalog		

3.4.2. UC Get AI Recommendations

ID and Name	UC29 - Get AI Recommendations		
Created by	Bùi Anh Linh		
Primary Actor	Customer	Secondary Actor	AI Recommendation Engine
Description	Users receive personalized product suggestions based on preferences and behavior.		
Preconditions	1. User logged in 2. Has completed style preferences or browsing history		

Main Sequence	1. User views For You page or homepage 2. System analyzes user profile and behavior 3. AI generates product recommendations 4. System displays personalized feed 5. User browses and clicks products
Alternative Sequences	2a. New user with no history: Show trending products + onboarding survey prompt
Postconditions	User sees relevant products matching their style. Higher engagement and conversion.
Business Rule	- Update recommendations daily - Mix styles to encourage discovery - Avoid showing out-of-stock items

3.4.3. UC View Smart Wardrobe

ID and Name	UC30 - View Smart Wardrobe		
Created by	Bùi Anh Linh		
Primary Actor	Customer	Secondary Actor	AI Styling Engine
Description	Users see outfit combinations from their purchased items.		
Preconditions	User has completed at least 2 orders.		
Main Sequence	1. User opens Smart Wardrobe 2. System displays purchased items as digital closet 3. AI suggests outfit combinations 4. User views styled looks with owned pieces		
Alternative Sequences	1a. < 2 orders: Show placeholder and encourage first purchase		
Postconditions	User inspired to wear existing items in new ways. May purchase complementary pieces.		
Business Rule	- Update wardrobe after each order - Suggest complementary products to complete looks		

3.4.4. UC Set Style Preferences

ID and Name	UC31 - Set Style Preferences
Created by	Bùi Anh Linh

Primary Actor	Customer	Secondary Actor	None
Description	Users define fashion style preferences during onboarding or in settings.		
Preconditions	User has account (new or existing).		
Main Sequence	1. System shows style quiz with images 2. User selects preferred styles (Y2K, Minimalist, Streetwear, etc.) 3. User sets budget range 4. User selects favorite brands (if any) 5. System saves preferences 6. AI uses data for personalization		
Alternative Sequences	None		
Postconditions	User preferences stored. AI recommendations personalized accordingly.		
Business Rule	- Allow multiple style selections - Preferences editable anytime in settings		

3.5. Social & Community

3.5.1. UC Post OOTD

ID and Name	UC32 - Post OOTD		
Created by	Bùi Anh Linh		
Primary Actor	Customer	Secondary Actor	Content Moderation System
Description	Users post outfit photos with product tags to share with community.		
Preconditions	1. User logged in 2. Accepted Terms of Service		
Main Sequence	1. User clicks Post OOTD 2. User uploads photo(s) 3. User tags products in photo 4. User adds caption (optional) 5. User clicks Post 6. System publishes OOTD to feed		
Alternative Sequences	2a. Image > 10MB: Compress automatically 5a. Inappropriate content detected: Block and warn user		
Postconditions	OOTD visible in community feed. Tagged products linked. User may receive likes/comments.		

Business Rule	- Max 5 photos per post - Auto-moderate for inappropriate content - Tag max 10 products
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3.5.2. UC Like OOTD Post

ID and Name	UC33 - Like OOTD Post		
Created by	Bùi Anh Linh		
Primary Actor	Customer	Secondary Actor	None
Description	Users like other users' OOTD posts.		
Preconditions	1. User logged in 2. Viewing OOTD feed		
Main Sequence	1. User taps heart icon on post 2. System increments like count 3. System notifies post author		
Alternative Sequences	1a. Already liked: Unlike post		
Postconditions	Like recorded. Post author notified.		
Business Rule	- One like per user per post - Like count visible to all		

3.5.3. UC Comment on OOTD

ID and Name	UC34 - Comment on OOTD		
Created by	Bùi Anh Linh		
Primary Actor	Customer	Secondary Actor	Content Moderation System
Description	Users comment on OOTD posts.		
Preconditions	User logged in.		
Main Sequence	1. User clicks comment icon 2. User types comment 3. User submits 4. System posts comment 5. System notifies post author		

Alternative Sequences	3a. Spam detected: Block comment and warn 3b. Empty comment: Show error
Postconditions	Comment visible under post. Author notified.
Business Rule	- Auto-moderate profanity - Allow emoji - Max 500 characters

3.5.4. UC Follow Brand

ID and Name	UC35 - Follow Brand		
Created by	Bùi Anh Linh		
Primary Actor	Customer	Secondary Actor	None
Description	Users follow favorite brands to receive updates on new products.		
Preconditions	User logged in.		
Main Sequence	1. User clicks Follow on brand profile 2. System adds brand to user's following list 3. System notifies brand of new follower		
Alternative Sequences	1a. Already following: Unfollow instead		
Postconditions	User receives notifications when brand posts new products/collections.		
Business Rule	- Unlimited follows - Brands can see follower count but not individual identities without consent		

3.5.5. UC Follow User

ID and Name	UC36 - Follow User		
Created by	Bùi Anh Linh		
Primary Actor	Customer	Secondary Actor	None
Description	Users follow other fashion enthusiasts.		
Preconditions	User logged in.		

Main Sequence	1. User clicks Follow on another user's profile 2. System creates follow relationship 3. System notifies followed user
Alternative Sequences	1a. Already following: Unfollow
Postconditions	User sees followed user's OOTD posts in feed.
Business Rule	- Public profiles only - Users can make profile private in settings

3.5.6. UC Submit Product Review

ID and Name	UC37 - Submit Product Review		
Created by	Bùi Anh Linh		
Primary Actor	Customer	Secondary Actor	None
Description	Users rate and review purchased products with photos.		
Preconditions	User has received order containing product.		
Main Sequence	1. System prompts review after delivery 2. User rates product (1-5 stars) 3. User writes review text 4. User uploads photos (optional) 5. User adds size/fit feedback 6. User submits review 7. System publishes review		
Alternative Sequences	6a. Inappropriate content: Flag for moderation		
Postconditions	Review visible on product page. Marked as 'Verified Purchase'.		
Business Rule	- Only verified purchases can review - Max 5 photos - Min 20 characters		

3.5.7. UC View Reviews

ID and Name	UC38 - View Reviews		
Created by	Bùi Anh Linh		
Primary Actor	Customer	Secondary Actor	None

Description	Users read verified reviews from other shoppers.
Preconditions	User viewing product detail page.
Main Sequence	1. User scrolls to Reviews section 2. System displays reviews with ratings 3. User filters by rating/size/date 4. User reads reviews and views photos
Alternative Sequences	None
Postconditions	User informed by peer experiences to make purchase decision.
Business Rule	- Show verified purchases first - Display average rating prominently

3.5.8. UC Report Content

ID and Name	UC39 - Report Content		
Created by	Bùi Anh Linh		
Primary Actor	Customer	Secondary Actor	Platform Admin
Description	Users report inappropriate content or users.		
Preconditions	1. User logged in 2. Viewing reportable content		
Main Sequence	1. User clicks Report button 2. System shows report reasons 3. User selects reason 4. User adds details (optional) 5. User submits report 6. System queues for admin review		
Alternative Sequences	None		
Postconditions	Report submitted to moderation queue. User receives confirmation.		
Business Rule	- Anonymous reports - Admin reviews within 24 hours - False reports may result in warnings		

3.5.9. UC Block User

ID and Name	UC40 - Block User
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Created by	Bùi Anh Linh		
Primary Actor	Customer	Secondary Actor	None
Description	Users block other users from interacting with them.		
Preconditions	User logged in.		
Main Sequence	1. User clicks Block on user profile 2. System confirms action 3. User confirms 4. System creates block relationship		
Alternative Sequences	None		
Postconditions	Blocked user cannot comment on user's posts, send messages, or view profile.		
Business Rule	- Block is permanent unless user unblocks - Blocked user not notified		

3.6. Brand Partner Features

3.6.1. UC Login to Brand Portal

ID and Name	UC41 - Login to Brand Portal		
Created by	Bùi Anh Linh		
Primary Actor	Local Brand Owner	Secondary Actor	Authentication Service
Description	Brand owners access partner portal with credentials.		
Preconditions	Brand account verified by admin.		
Main Sequence	1. Brand owner navigates to portal.wearwhere.vn 2. Brand owner enters credentials 3. System validates and creates session 4. System displays brand dashboard		
Alternative Sequences	3a. Wrong credentials: Show error and retry		
Postconditions	Brand owner logged into portal. Can manage products and orders.		
Business Rule	- Separate authentication from customer accounts - Multi-user support per brand		

3.6.2. UC Manage Product Catalog

ID and Name	UC42 - Manage Product Catalog		
Created by	Bùi Anh Linh		
Primary Actor	Local Brand Owner	Secondary Actor	Platform Admin
Description	Brands add, edit, or remove products from catalog.		
Preconditions	Brand logged into portal.		
Main Sequence	ADD: 1. Click Add Product → Fill form → Upload images → Save EDIT: 1. Select product → Modify fields → Update DELETE: 1. Select product → Delete → Confirm		
Alternative Sequences	Validation errors: Highlight missing/invalid fields		
Postconditions	Product catalog updated. Changes reflected on customer-facing platform within 30 seconds.		
Business Rule	- Max 500 products (Standard), 2000 (Premium) - Min 3 images per product - Price 10K-50M VND		

3.6.3. UC Upload Product Images

ID and Name	UC43 - Upload Product Images		
Created by	Bùi Anh Linh		
Primary Actor	Local Brand Owner	Secondary Actor	Cloud Storage
Description	Brands upload high-quality product photos.		
Preconditions	Adding or editing product.		
Main Sequence	1. Brand clicks upload area 2. Brand selects images (drag-drop or file picker) 3. System validates image quality 4. System uploads and processes images 5. System displays thumbnails		
Alternative Sequences	3a. Low quality (< 800px): Reject with error 3b. Wrong format: Accept JPG/PNG only		
Postconditions	Images uploaded and associated with product.		

Business Rule	- Min 800x800px - Max 5MB per image - Max 10 images per product - JPG/PNG only
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3.6.4. UC Set Product Pricing

ID and Name	UC44 - Set Product Pricing		
Created by	Bùi Anh Linh		
Primary Actor	Local Brand Owner	Secondary Actor	None
Description	Brands set prices and manage inventory per variant.		
Preconditions	Creating or editing product.		
Main Sequence	1. Brand enters base price 2. Brand sets compare-at price (optional) 3. Brand defines variants (size/color) 4. Brand sets stock quantity per variant 5. System saves pricing data		
Alternative Sequences	5a. Price out of range (< 10K or > 50M): Show warning		
Postconditions	Pricing active. Stock tracked per variant. Discount % calculated automatically.		
Business Rule	- Price range: 10K-50M VND - Stock updates real-time - Out-of-stock shows 'Notify' button		

3.6.5. UC View Orders

ID and Name	UC45 - View Orders		
Created by	Bùi Anh Linh		
Primary Actor	Local Brand Owner	Secondary Actor	None
Description	Brands view incoming orders to prepare shipment.		
Preconditions	Brand logged into portal.		

Main Sequence	1. Brand navigates to Orders section 2. System displays order list (newest first) 3. Brand filters by status (Pending/Confirmed/Shipped) 4. Brand clicks order to view details
Alternative Sequences	None
Postconditions	Brand aware of orders needing fulfillment.
Business Rule	- Show customer name, address, products, payment status - Orders expire if not confirmed in 24h

3.6.6. UC Confirm Order

ID and Name	UC46 - Confirm Order		
Created by	Bùi Anh Linh		
Primary Actor	Local Brand Owner	Secondary Actor	Customer
Description	Brands accept orders and begin preparation.		
Preconditions	Order status is Pending.		
Main Sequence	1. Brand views order details 2. Brand verifies stock availability 3. Brand clicks Confirm Order 4. System updates status to Confirmed 5. System notifies customer		
Alternative Sequences	2a. Out of stock: Brand cancels order with reason		
Postconditions	Order confirmed. Customer notified. Brand must ship within 3 days.		
Business Rule	- Must confirm within 24h or auto-cancel - Commission charged upon confirmation		

3.6.7. UC Update Order Status

ID and Name	UC47 - Update Order Status		
Created by	Bùi Anh Linh		
Primary Actor	Local Brand Owner	Secondary Actor	Customer
Description	Brands update shipping and delivery status.		

Preconditions	Order confirmed.
Main Sequence	1. Brand prepares package 2. Brand marks order as Preparing 3. Brand ships order 4. Brand updates status to Shipped (with tracking) 5. Brand confirms delivery 6. System updates customer
Alternative Sequences	None
Postconditions	Order status reflects reality. Customer informed at each stage.
Business Rule	- Status updates trigger customer notifications - Tracking number required for Shipped status

3.6.8. UC Manage Reviews

ID and Name	UC48 - Manage Reviews		
Created by	Bùi Anh Linh		
Primary Actor	Local Brand Owner	Secondary Actor	None
Description	Brands respond to customer reviews.		
Preconditions	Product has reviews.		
Main Sequence	1. Brand views product reviews 2. Brand clicks Reply on review 3. Brand writes response 4. Brand submits reply 5. System publishes reply under review		
Alternative Sequences	None		
Postconditions	Brand response visible. Shows engagement and customer service.		
Business Rule	- One reply per review - Cannot edit/delete customer reviews - Professional tone expected		

3.6.9. UC View Analytics Dashboard

ID and Name	UC49 - View Analytics Dashboard
Created by	Bùi Anh Linh

Primary Actor	Local Brand Owner	Secondary Actor	None
Description	Brands view sales, traffic, and customer insights.		
Preconditions	Brand logged into portal.		
Main Sequence	1. Brand navigates to Analytics 2. System displays dashboard with KPIs 3. Brand views metrics: GMV, orders, traffic, conversion 4. Brand filters by date range 5. Brand exports reports if needed		
Alternative Sequences	None		
Postconditions	Brand understands performance and can optimize strategy.		
Business Rule	- Update metrics daily - Show trends vs. previous period - Include customer demographics		

3.6.10. UC Post Brand Content

ID and Name	UC50 - Post Brand Content		
Created by	Bùi Anh Linh		
Primary Actor	Local Brand Owner	Secondary Actor	None
Description	Brands publish new collections and brand stories.		
Preconditions	Brand logged into portal.		
Main Sequence	1. Brand clicks Create Post 2. Brand selects post type (New Collection/Story/Event) 3. Brand uploads media and writes content 4. Brand publishes post 5. System notifies brand followers		
Alternative Sequences	None		
Postconditions	Content visible on brand profile. Followers notified.		
Business Rule	- Posts appear in follower feeds - Can tag products for direct shopping		

3.6.11. UC Manage Store Info

ID and Name	UC51 - Manage Store Info		
Created by	Bùi Anh Linh		
Primary Actor	Local Brand Owner	Secondary Actor	Goong Maps API
Description	Brands update store locations, hours, and contact info.		
Preconditions	Brand logged into portal.		
Main Sequence	1. Brand navigates to Store Settings 2. Brand edits address/hours/phone 3. Brand updates store photos 4. Brand saves changes 5. System updates map and store listings		
Alternative Sequences	4a. Invalid address: Show validation error		
Postconditions	Store info updated. Customers see accurate data on map and search.		
Business Rule	- Store hours affect open/closed status display - Photos improve store appeal		

3.7. Admin Features

3.7.1. UC Login to Admin CMS

ID and Name	UC52 - Login to Admin CMS		
Created by	Bùi Anh Linh		
Primary Actor	Platform Admin	Secondary Actor	Authentication Service
Description	Admins access CMS with admin credentials.		
Preconditions	User has admin role.		
Main Sequence	1. Admin navigates to admin.wearwhere.vn 2. Admin enters credentials 3. System validates admin role 4. System displays CMS dashboard		
Alternative Sequences	3a. Not admin: Deny access		
Postconditions	Admin logged in with full system access.		

Business Rule	- 2FA required for admin accounts - Audit log all admin actions
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3.7.2. UC Verify Brand Applications

ID and Name	UC53 - Verify Brand Applications		
Created by	Bùi Anh Linh		
Primary Actor	Platform Admin	Secondary Actor	Local Brand Owner
Description	Admins approve or reject new brand registrations.		
Preconditions	Brand submitted application.		
Main Sequence	1. Admin views pending applications 2. Admin reviews brand info, portfolio, social media 3. Admin approves or rejects with reason 4. System notifies brand of decision		
Alternative Sequences	3a. Needs more info: Admin requests additional details		
Postconditions	Approved brands can access portal. Rejected brands notified with reason.		
Business Rule	- Verify design authenticity, quality standards - Review within 3 business days		

3.7.3. UC Manage Users

ID and Name	UC54 - Manage Users		
Created by	Bùi Anh Linh		
Primary Actor	Platform Admin	Secondary Actor	None
Description	Admins view, suspend, or delete user accounts.		
Preconditions	Admin logged in.		
Main Sequence	1. Admin searches for user 2. Admin views user profile and activity 3. Admin takes action (suspend/delete/warn) 4. System executes action and logs		
Alternative Sequences	None		
Postconditions	User account modified. User notified if suspended/deleted.		

Business Rule	- Deletion permanent after 90 days - Suspension temporary (7/30/90 days) - Log all actions
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3.7.4. UC Moderate Content

ID and Name	UC55 - Moderate Content		
Created by	Bùi Anh Linh		
Primary Actor	Platform Admin	Secondary Actor	None
Description	Admins review and remove inappropriate UGC.		
Preconditions	Content flagged by auto-filter or user report.		
Main Sequence	1. Admin views moderation queue 2. Admin reviews flagged content 3. Admin approves or removes content 4. If removed: Admin warns/suspends user 5. System updates content status		
Alternative Sequences	None		
Postconditions	Inappropriate content removed. User warned or suspended if severe.		
Business Rule	- Review reports within 24h - Three violations = account suspension		

3.7.5. UC Manage Product Listings

ID and Name	UC56 - Manage Product Listings		
Created by	Bùi Anh Linh		
Primary Actor	Platform Admin	Secondary Actor	Local Brand Owner
Description	Admins review and remove policy-violating products.		
Preconditions	Product flagged or reported.		
Main Sequence	1. Admin views flagged products 2. Admin reviews product details 3. Admin approves or removes product 4. If removed: Admin notifies brand with reason		

Alternative Sequences	None
Postconditions	Policy-compliant products remain. Violations removed.
Business Rule	- No counterfeit, prohibited items, inappropriate content - Brand warned on violations

3.7.6. UC Handle Reports

ID and Name	UC57 - Handle Reports		
Created by	Bùi Anh Linh		
Primary Actor	Platform Admin	Secondary Actor	Customer
Description	Admins process user reports and take actions.		
Preconditions	User submitted report.		
Main Sequence	1. Admin views report queue 2. Admin investigates reported content/user 3. Admin takes action (remove content, warn user, etc.) 4. Admin closes report with resolution 5. System notifies reporter		
Alternative Sequences	3a. False report: Warn reporter		
Postconditions	Report resolved. Appropriate action taken.		
Business Rule	- Review within 24h - Three false reports = warning to reporter		

3.7.7. UC Configure Platform Settings

ID and Name	UC58 - Configure Platform Settings		
Created by	Bùi Anh Linh		
Primary Actor	Platform Admin	Secondary Actor	None
Description	Admins adjust commission rates, fees, and policies.		
Preconditions	Admin logged in with super admin role.		

Main Sequence	1. Admin navigates to Settings 2. Admin modifies configuration (commission %, policies) 3. Admin saves changes 4. System applies new settings
Alternative Sequences	None
Postconditions	Platform operates with updated parameters.
Business Rule	- Commission changes apply to new orders only - Log all config changes

3.7.8. UC Manage News & Editorial

ID and Name	UC59 - Manage News & Editorial		
Created by	Bùi Anh Linh		
Primary Actor	Platform Admin	Secondary Actor	None
Description	Admins publish fashion news and trend articles.		
Preconditions	Admin logged in.		
Main Sequence	1. Admin creates new article 2. Admin writes content and uploads images 3. Admin sets publish date 4. Admin publishes article 5. System displays in news feed		
Alternative Sequences	None		
Postconditions	Article visible to users. Drives engagement and platform value.		
Business Rule	- Editorial content supports brand discovery and fashion education		

3.7.9. UC View System Analytics

ID and Name	UC60 - View System Analytics		
Created by	Bùi Anh Linh		
Primary Actor	Platform Admin	Secondary Actor	None
Description	Admins view GMV, MAU, orders, and platform KPIs.		
Preconditions	Admin logged in.		

Main Sequence	1. Admin navigates to Analytics dashboard 2. System displays key metrics and charts 3. Admin filters by date/segment 4. Admin exports data if needed
Alternative Sequences	None
Postconditions	Admin understands platform health and growth.
Business Rule	- Real-time data for critical metrics - Daily updates for others

3.7.10. UC Generate Reports

ID and Name	UC61 - Generate Reports		
Created by	Bùi Anh Linh		
Primary Actor	Platform Admin	Secondary Actor	None
Description	Admins export revenue, user, and transaction reports.		
Preconditions	Admin logged in.		
Main Sequence	1. Admin selects report type 2. Admin sets date range and filters 3. Admin clicks Generate 4. System compiles report 5. System provides download link		
Alternative Sequences	4a. Large dataset: Queue report and email when ready		
Postconditions	Admin has detailed report for analysis or compliance.		
Business Rule	- Support CSV, Excel, PDF formats - Include all necessary accounting data		

3.7.11. UC Monitor Transactions

ID and Name	UC62 - Monitor Transactions		
Created by	Bùi Anh Linh		
Primary Actor	Platform Admin	Secondary Actor	Payment Gateway
Description	Admins track payment status and resolve issues.		
Preconditions	Admin logged in.		

Main Sequence	1. Admin views transaction log 2. Admin filters by status/payment method 3. Admin investigates failed transactions 4. Admin resolves issues or contacts support 5. System logs resolution
Alternative Sequences	None
Postconditions	Payment issues resolved. Transaction data accurate.
Business Rule	- Track all payments (COD, Momo, VNPay) - Flag suspicious activity for review

4. Non-Functional Requirements

4.1 External Interfaces

4.1.1 User Interface (UI)

Guest UI:

- Homepage with featured brands, trending products, OOTD feed
- Product catalog with search and filters
- Login/registration forms with social login options (Google, Facebook, Apple Sign-In)
- Password recovery via email/SMS OTP

Customer UI (Web & Mobile):

- Personalized homepage with AI recommendations
- Shopping cart and checkout flow with multiple payment methods
- Order tracking and history dashboard
- Store finder with GPS integration and Goong Maps
- AI Stylist chatbot interface
- Smart Wardrobe with outfit suggestions
- OOTD feed with post, like, comment capabilities
- Profile management and settings

Brand Partner Portal (Web):

- Product catalog management (CRUD operations)
- Order management dashboard with status updates
- Analytics dashboard with sales, traffic, customer insights
- Brand content publishing interface
- Review management and response system

Admin CMS (Web):

- Brand verification and approval workflow
- User and content moderation dashboard
- Product listing oversight
- Platform configuration (commission rates, policies)
- System analytics with GMV, MAU, order metrics
- Report generation and transaction monitoring

Design Requirements:

- Fully responsive across desktop, tablet, mobile

- Accessible (WCAG 2.1 AA compliance)
- Intuitive navigation with clear CTAs
- Real-time validation on forms
- Loading states and error handling
- Consistent design system across platforms

4.1.2 Application Programming Interface (API)

Authentication API:

- POST [/auth/register](#) - Create account (email/phone/social)
- POST [/auth/login](#) - User login with JWT token
- POST [/auth/logout](#) - Invalidate session
- POST [/auth/forgot-password](#) - Request OTP
- POST [/auth/reset-password](#) - Update password with OTP
- POST [/auth/refresh-token](#) - Refresh JWT token

User API:

- GET [/users/profile](#) - Get user profile
- PUT [/users/profile](#) - Update profile
- PUT [/users/password](#) - Change password
- DELETE [/users/account](#) - Delete account
- PUT [/users/preferences](#) - Update style preferences

Product API:

- GET [/products](#) - List products with filters
- GET [/products/:id](#) - Product details
- GET [/products/search](#) - Search products
- GET [/brands/:id/products](#) - Products by brand

Shopping API:

- POST [/cart](#) - Add to cart
- GET [/cart](#) - Get cart items
- PUT [/cart/:id](#) - Update cart item
- DELETE [/cart/:id](#) - Remove from cart
- POST [/orders](#) - Place order
- GET [/orders](#) - Order history
- GET [/orders/:id](#) - Order details
- PUT [/orders/:id/cancel](#) - Cancel order

Store API:

- GET [/stores/nearby](#) - Find stores by GPS (requires lat/lng)
- GET [/stores/search](#) - Search stores by area
- GET [/stores/:id](#) - Store details

AI API:

- POST [/ai/chat](#) - AI Stylist conversation
- GET [/ai/recommendations](#) - Personalized product suggestions
- GET [/ai/wardrobe](#) - Smart wardrobe outfits

Social API:

- POST [/ootd](#) - Create OOTD post

- GET /ootd - OOTD feed
- POST /ootd/:id/like - Like post
- POST /ootd/:id/comment - Comment on post
- POST /brands/:id/follow - Follow brand
- POST /products/:id/review - Submit review

Brand Portal API:

- POST /brand/products - Create product
- PUT /brand/products/:id - Update product
- DELETE /brand/products/:id - Delete product
- GET /brand/orders - View orders
- PUT /brand/orders/:id/confirm - Confirm order
- PUT /brand/orders/:id/status - Update order status
- GET /brand/analytics - View analytics

Admin API:

- GET /admin/brands/pending - Pending brand applications
- PUT /admin/brands/:id/verify - Approve/reject brand
- GET /admin/users - List users
- PUT /admin/users/:id/suspend - Suspend user
- GET /admin/content/flagged - Moderation queue
- DELETE /admin/content/:id - Remove content
- GET /admin/analytics - System metrics
- POST /admin/reports - Generate reports

API Standards:

- RESTful architecture
- JSON request/response format
- JWT authentication with role-based authorization
- Rate limiting: 100 requests/minute per user
- Pagination for list endpoints (limit/offset)
- Versioning: /api/v1/...
- Error responses with standard HTTP status codes

4.1.3 External Systems

Payment Gateways:

- **Momo** - E-wallet payment integration
- **VNPay** - Bank card payment integration
- PCI-DSS compliant transaction processing
- Webhook callbacks for payment confirmation
- Refund support for order cancellations

Maps & Location:

- **Goong Maps API** - Vietnam-specific mapping
- Geocoding for store addresses
- Turn-by-turn navigation
- Distance calculation
- Store location markers

AI Services:

- **Claude API (Anthropic)** - AI Stylist chatbot
- Natural language understanding for fashion queries
- Product recommendation generation
- Token usage tracking and quota management

Cloud Services (Google Cloud Platform):

- **Cloud Storage** - Product images, user avatars, OOTD photos
- **Cloud Run** - Backend hosting
- **Cloud SQL** - PostgreSQL database
- **Cloud CDN** - Static asset delivery

Communication:

- **Email Service (SendGrid/Firebase)** - Transactional emails (order confirmations, password reset)
- **SMS Service** - OTP delivery for authentication

Analytics:

- **Google Analytics** - User behavior tracking
- **Sentry** - Error monitoring and logging

4.1.4 Database

PostgreSQL (Primary Database):

- Users (customers, brand owners, admins)
- Products and variants (size, color, stock)
- Orders and order items
- Brands and stores
- Reviews and ratings
- OOTD posts, likes, comments
- Transactions and payment history

Redis (Cache & Session Store):

- Session tokens
- Shopping cart data
- API rate limiting
- Frequently accessed product data
- Real-time stock counters

4.2 Quality Attributes

4.2.1 Usability

- **Intuitive Navigation:** Max 3 clicks to reach any product
- **Onboarding:** New users complete style quiz in under 2 minutes
- **AI Stylist:** Clear prompts guide user questions
- **Mobile-First:** Optimized for touch interactions
- **Accessibility:** Screen reader compatible, keyboard navigation
- **Multilingual:** Vietnamese and English interface
- **Error Prevention:** Clear validation messages before submission
- **Help & Support:** In-app help center and chatbot

4.2.2 Reliability

- **Uptime:** 99.9% availability (max 8.76 hours downtime/year)
- **Data Consistency:** ACID transactions for orders and payments
- **Fault Tolerance:** Graceful degradation if AI services unavailable
- **Backup:** Daily automated backups with 30-day retention
- **Disaster Recovery:** Recovery Time Objective (RTO) < 4 hours
- **Error Handling:** Retry logic for payment gateway timeouts
- **Monitoring:** Real-time alerts for system failures

4.2.3 Performance

- **Page Load:** < 2 seconds for product pages
- **API Response:** < 500ms for 95% of requests
- **AI Stylist:** Response within 3 seconds
- **Search:** Results in < 1 second
- **Image Load:** Progressive loading with lazy loading for below-fold images
- **Database:** Indexed queries, < 100ms query time
- **Caching:** 80% cache hit rate for product catalog
- **Concurrent Users:** Support 10,000 simultaneous users
- **CDN:** Static assets served from edge locations

4.2.4 Security

Authentication & Authorization:

- bcrypt password hashing (cost factor 12)
- JWT tokens with 30-day expiration
- Role-based access control (Guest, Customer, Brand, Admin)
- Apple Sign-In required on iOS (App Store compliance)
- 2FA for admin accounts

Data Protection:

- HTTPS/TLS 1.3 for all connections
- Encryption at rest for sensitive data
- PCI-DSS compliant payment processing (no card data stored)
- GDPR compliant (data deletion within 90 days)

Attack Prevention:

- SQL injection protection (parameterized queries)
- XSS prevention (input sanitization, CSP headers)
- CSRF tokens for state-changing operations
- Rate limiting to prevent brute force (5 login attempts = 15min lockout)
- Content Security Policy headers
- DDoS protection via Cloud CDN

Compliance:

- App Store Guidelines 4.8 (Apple Sign-In)
- Google Play Data Safety requirements
- User account deletion capability (GDPR)
- Privacy Policy and Terms of Service acceptance

- Content moderation for UGC

Audit & Monitoring:

- All admin actions logged with timestamps
- Failed login attempts tracked
- Suspicious transaction flagging
- Real-time security alerts via Sentry